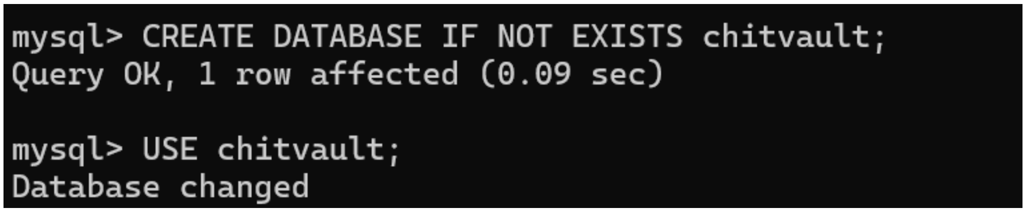
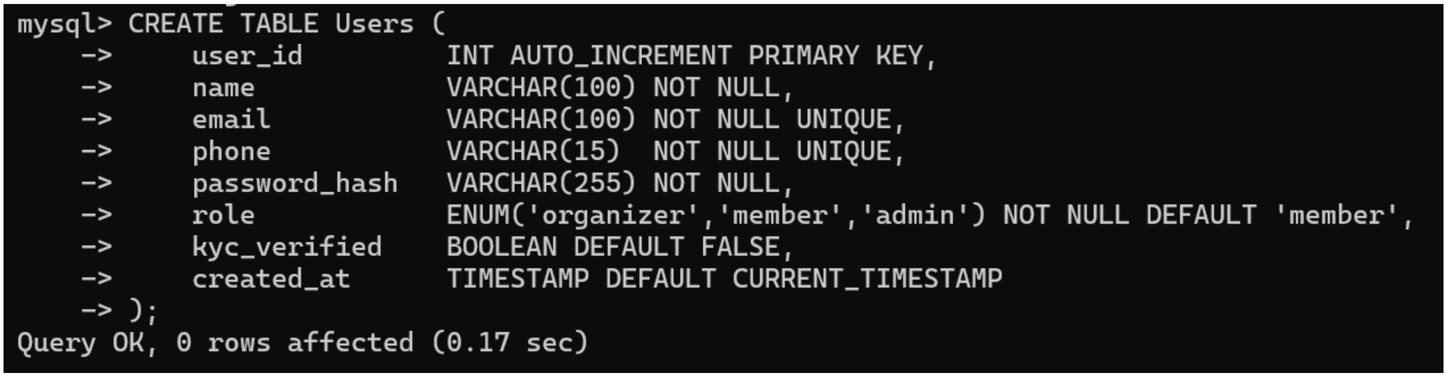
# A Database-Driven Chit Fund Management and Auction System

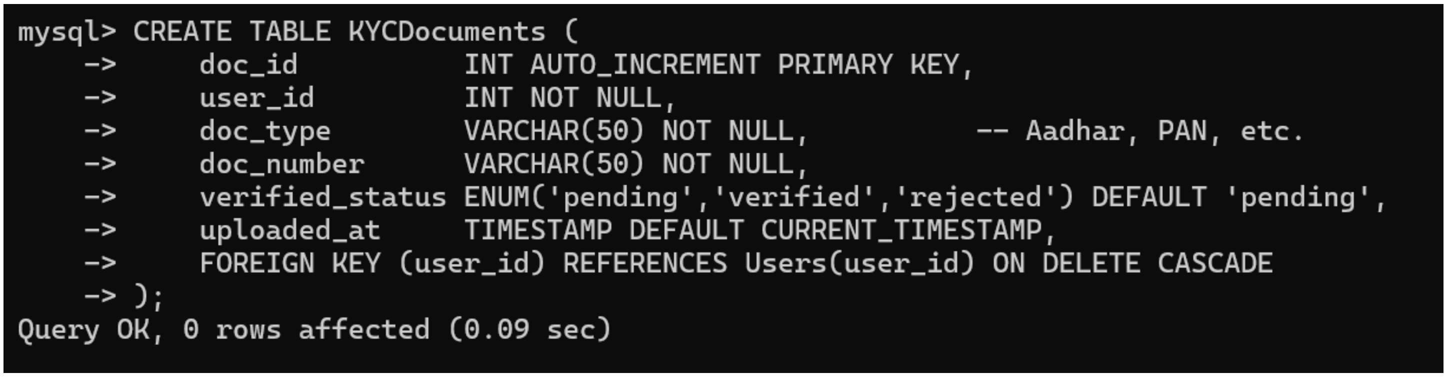
## DDL



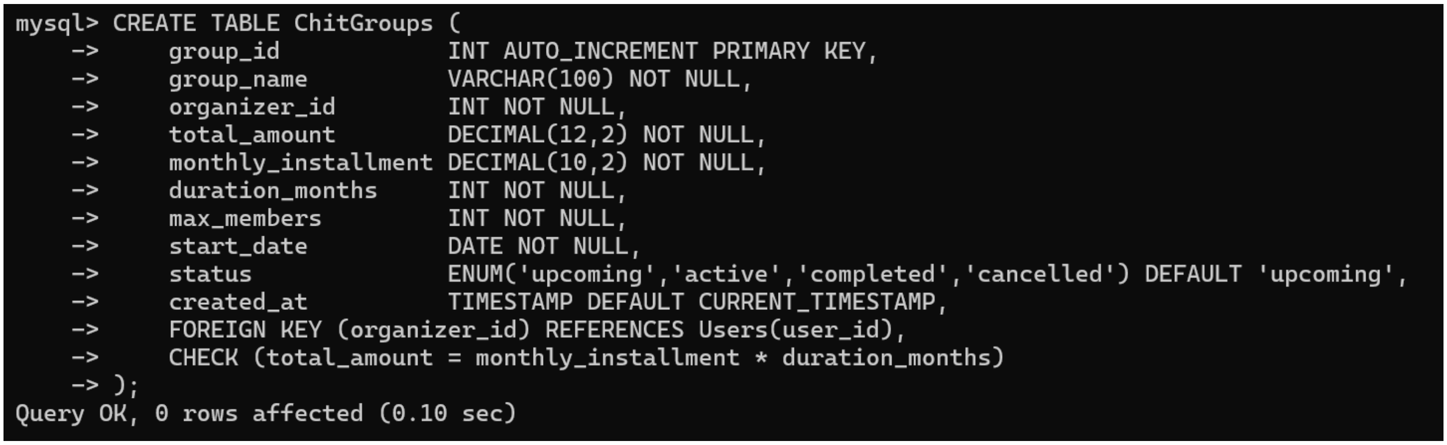
```
mysql> CREATE DATABASE IF NOT EXISTS chitvault;
Query OK, 1 row affected (0.09 sec)

mysql> USE chitvault;
Database changed
```



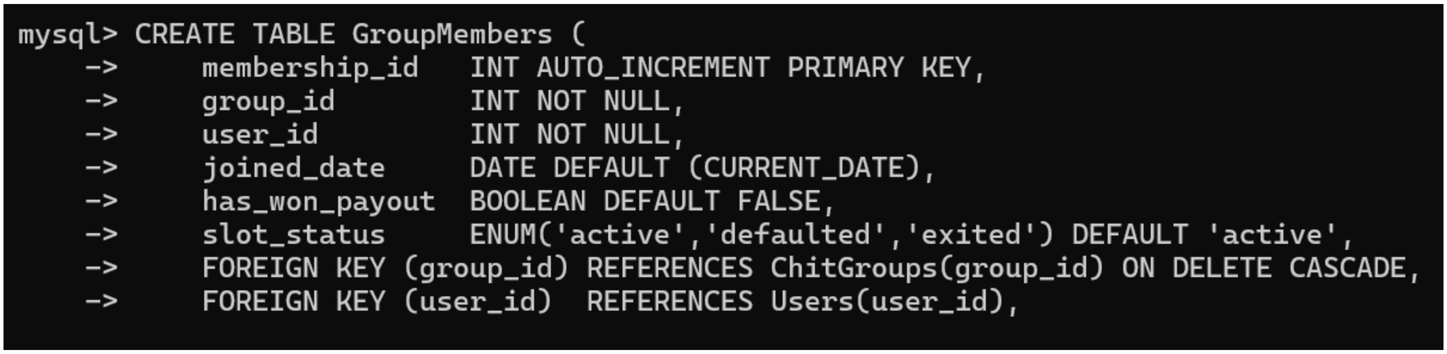
```
mysql> CREATE TABLE Users (
    ->     user_id         INT AUTO_INCREMENT PRIMARY KEY,
    ->     name            VARCHAR(100) NOT NULL,
    ->     email           VARCHAR(100) NOT NULL UNIQUE,
    ->     phone           VARCHAR(15)  NOT NULL UNIQUE,
    ->     password_hash   VARCHAR(255) NOT NULL,
    ->     role            ENUM('organizer','member','admin') NOT NULL DEFAULT 'member',
    ->     kyc_verified    BOOLEAN DEFAULT FALSE,
    ->     created_at      TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.17 sec)
```



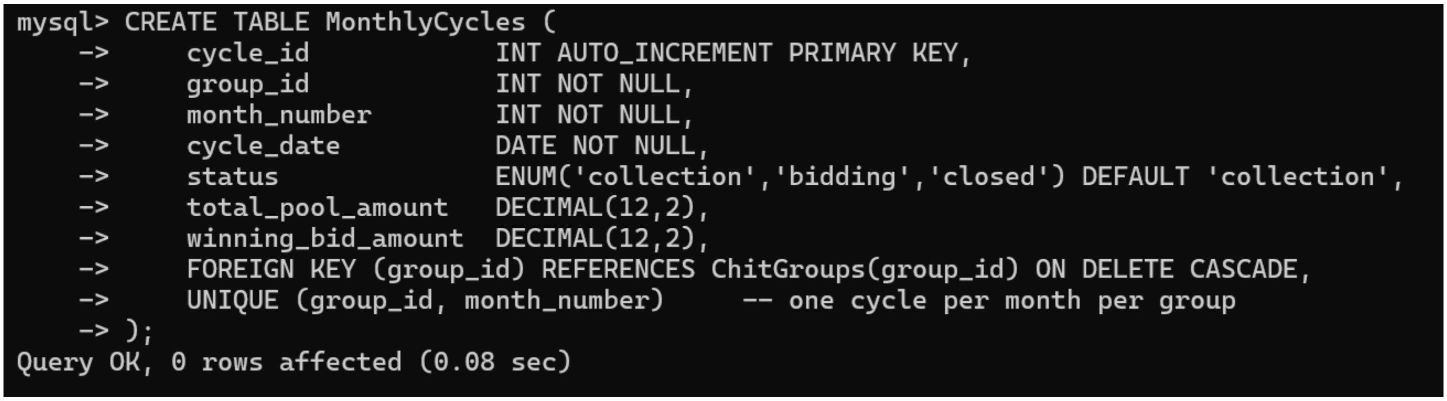
```
mysql> CREATE TABLE KYCDocuments (
    ->     doc_id          INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id         INT NOT NULL,
    ->     doc_type        VARCHAR(50) NOT NULL,          -- Aadhar, PAN, etc.
    ->     doc_number      VARCHAR(50) NOT NULL,
    ->     verified_status ENUM('pending','verified','rejected') DEFAULT 'pending',
    ->     uploaded_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (user_id) REFERENCES Users(user_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.09 sec)
```



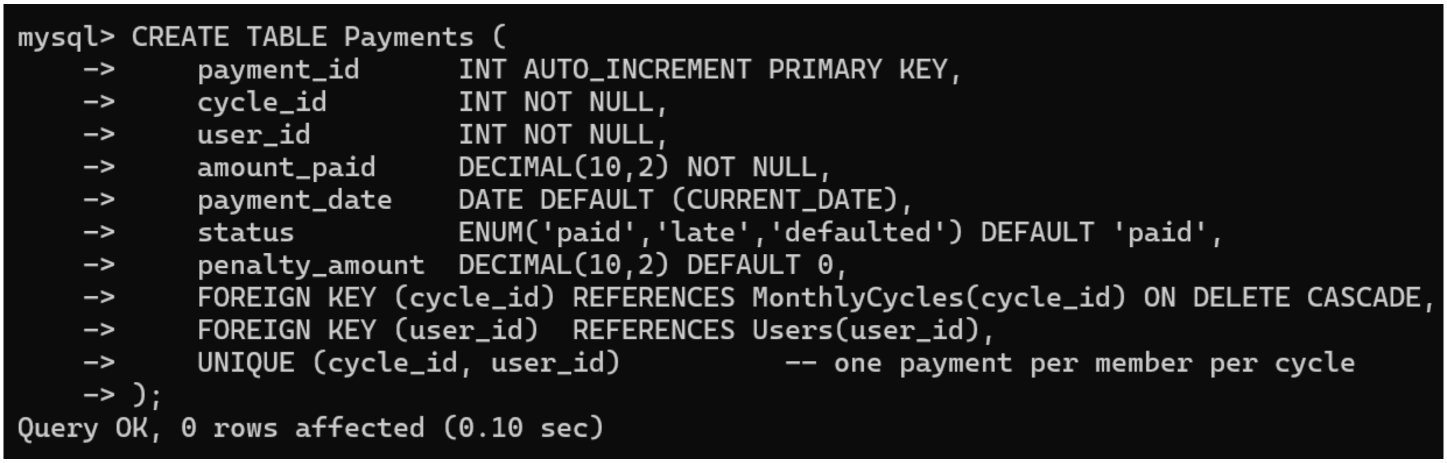
```
mysql> CREATE TABLE ChitGroups (
    ->     group_id            INT AUTO_INCREMENT PRIMARY KEY,
    ->     group_name          VARCHAR(100) NOT NULL,
    ->     organizer_id        INT NOT NULL,
    ->     total_amount        DECIMAL(12,2) NOT NULL,
    ->     monthly_installment DECIMAL(10,2) NOT NULL,
    ->     duration_months     INT NOT NULL,
    ->     max_members         INT NOT NULL,
    ->     start_date          DATE NOT NULL,
    ->     status              ENUM('upcoming','active','completed','cancelled') DEFAULT 'upcoming',
    ->     created_at          TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (organizer_id) REFERENCES Users(user_id),
    ->     CHECK (total_amount = monthly_installment * duration_months)
    -> );
Query OK, 0 rows affected (0.10 sec)
```



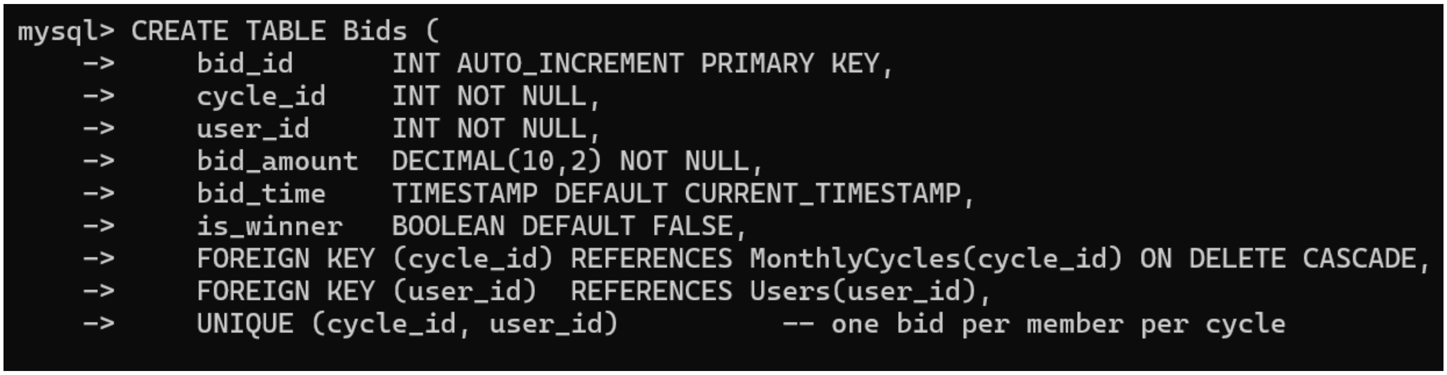
```
mysql> CREATE TABLE GroupMembers (
    ->     membership_id   INT AUTO_INCREMENT PRIMARY KEY,
    ->     group_id        INT NOT NULL,
    ->     user_id         INT NOT NULL,
    ->     joined_date     DATE DEFAULT (CURRENT_DATE),
    ->     has_won_payout  BOOLEAN DEFAULT FALSE,
    ->     slot_status     ENUM('active','defaulted','exited') DEFAULT 'active',
    ->     FOREIGN KEY (group_id) REFERENCES ChitGroups(group_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (user_id)  REFERENCES Users(user_id),
```



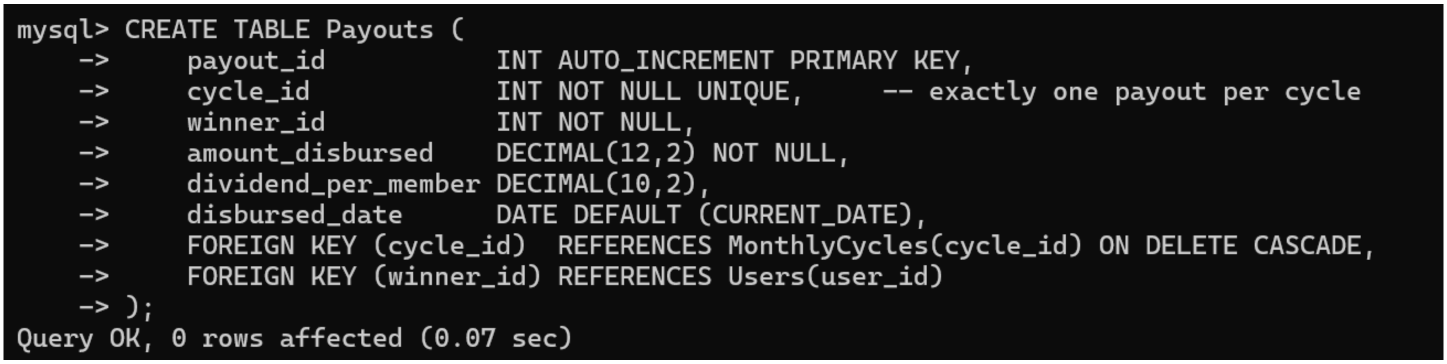
```
mysql> CREATE TABLE MonthlyCycles (
    ->     cycle_id            INT AUTO_INCREMENT PRIMARY KEY,
    ->     group_id            INT NOT NULL,
    ->     month_number        INT NOT NULL,
    ->     cycle_date          DATE NOT NULL,
    ->     status              ENUM('collection','bidding','closed') DEFAULT 'collection',
    ->     total_pool_amount   DECIMAL(12,2),
    ->     winning_bid_amount  DECIMAL(12,2),
    ->     FOREIGN KEY (group_id) REFERENCES ChitGroups(group_id) ON DELETE CASCADE,
    ->     UNIQUE (group_id, month_number)     -- one cycle per month per group
    -> );
Query OK, 0 rows affected (0.08 sec)
```



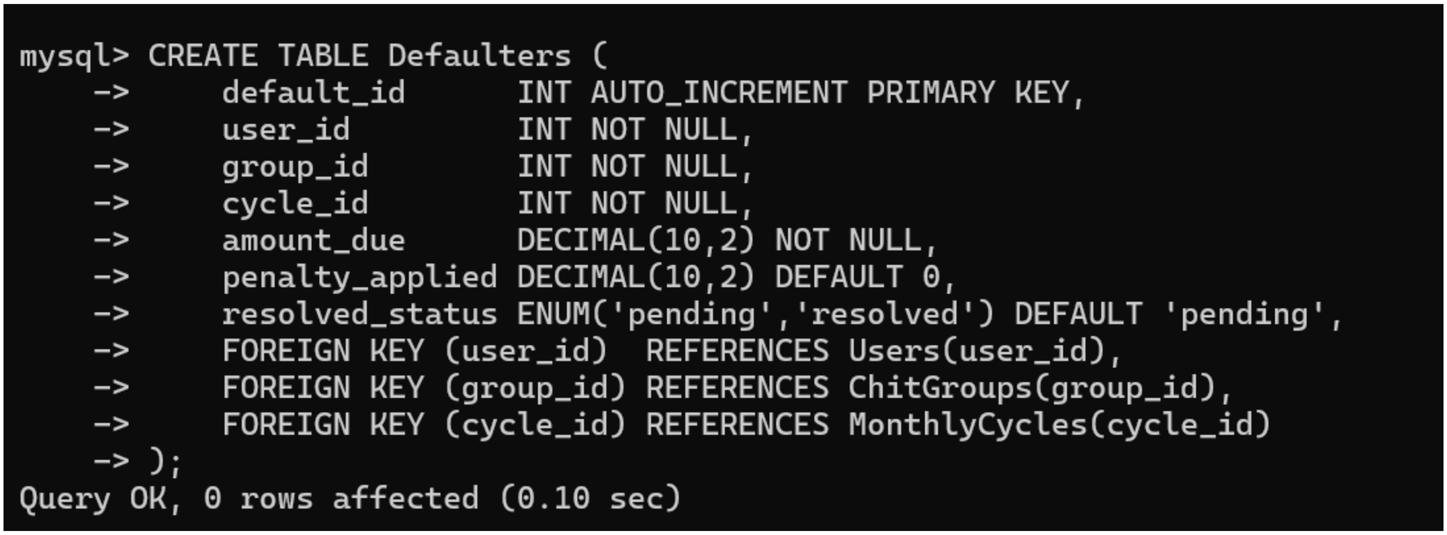
```
mysql> CREATE TABLE Payments (
    ->     payment_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     cycle_id        INT NOT NULL,
    ->     user_id         INT NOT NULL,
    ->     amount_paid     DECIMAL(10,2) NOT NULL,
    ->     payment_date    DATE DEFAULT (CURRENT_DATE),
    ->     status          ENUM('paid','late','defaulted') DEFAULT 'paid',
    ->     penalty_amount  DECIMAL(10,2) DEFAULT 0,
    ->     FOREIGN KEY (cycle_id) REFERENCES MonthlyCycles(cycle_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (user_id)  REFERENCES Users(user_id),
    ->     UNIQUE (cycle_id, user_id)          -- one payment per member per cycle
    -> );
Query OK, 0 rows affected (0.10 sec)
```



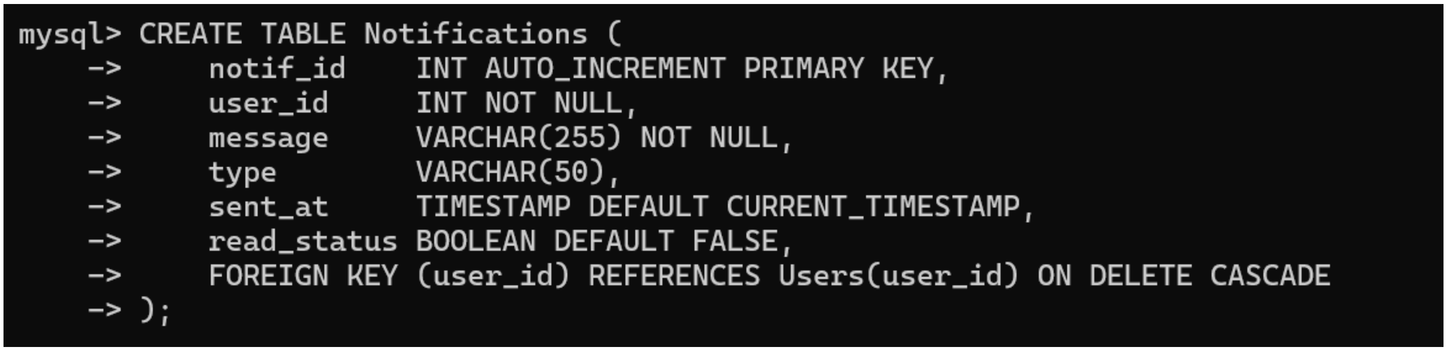
```
mysql> CREATE TABLE Bids (
    ->     bid_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     cycle_id    INT NOT NULL,
    ->     user_id     INT NOT NULL,
    ->     bid_amount  DECIMAL(10,2) NOT NULL,
    ->     bid_time    TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     is_winner   BOOLEAN DEFAULT FALSE,
    ->     FOREIGN KEY (cycle_id) REFERENCES MonthlyCycles(cycle_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (user_id)  REFERENCES Users(user_id),
    ->     UNIQUE (cycle_id, user_id)          -- one bid per member per cycle
```



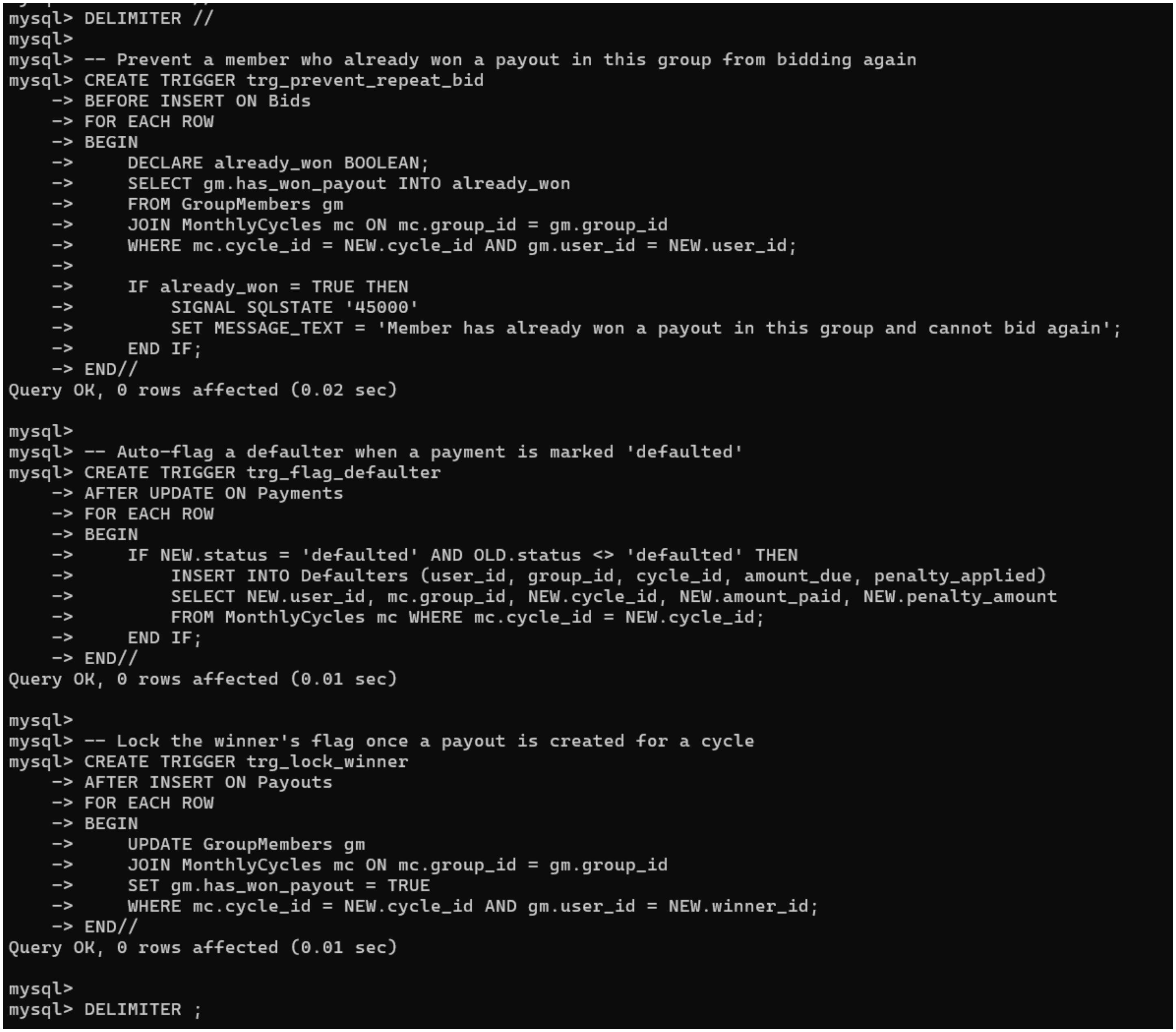
```
mysql> CREATE TABLE Payouts (
    ->     payout_id           INT AUTO_INCREMENT PRIMARY KEY,
    ->     cycle_id            INT NOT NULL UNIQUE,     -- exactly one payout per cycle
    ->     winner_id           INT NOT NULL,
    ->     amount_disbursed    DECIMAL(12,2) NOT NULL,
    ->     dividend_per_member DECIMAL(10,2),
    ->     disbursed_date      DATE DEFAULT (CURRENT_DATE),
    ->     FOREIGN KEY (cycle_id)  REFERENCES MonthlyCycles(cycle_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (winner_id) REFERENCES Users(user_id)
    -> );
Query OK, 0 rows affected (0.07 sec)
```



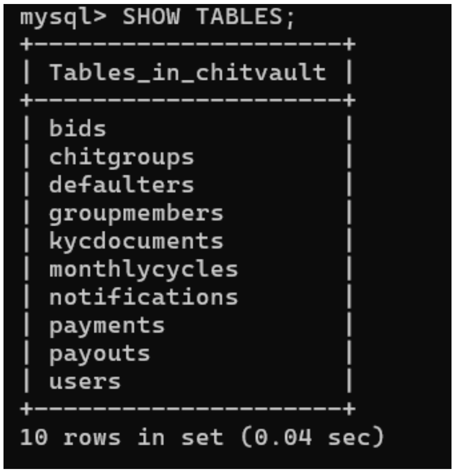
```
mysql> CREATE TABLE Defaulters (
    ->     default_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id         INT NOT NULL,
    ->     group_id        INT NOT NULL,
    ->     cycle_id        INT NOT NULL,
    ->     amount_due      DECIMAL(10,2) NOT NULL,
    ->     penalty_applied DECIMAL(10,2) DEFAULT 0,
    ->     resolved_status ENUM('pending','resolved') DEFAULT 'pending',
    ->     FOREIGN KEY (user_id)  REFERENCES Users(user_id),
    ->     FOREIGN KEY (group_id) REFERENCES ChitGroups(group_id),
    ->     FOREIGN KEY (cycle_id) REFERENCES MonthlyCycles(cycle_id)
    -> );
Query OK, 0 rows affected (0.10 sec)
```



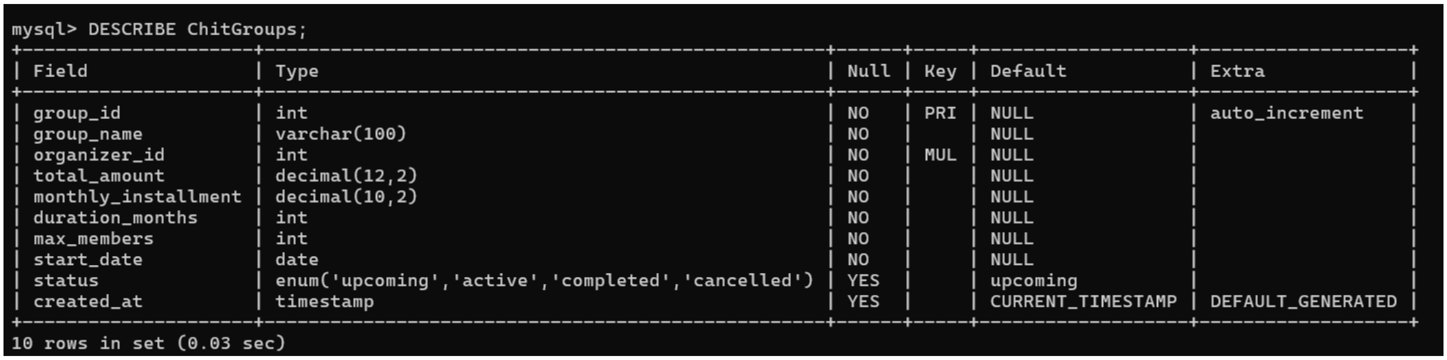
```
mysql> CREATE TABLE Notifications (
    ->     notif_id    INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id     INT NOT NULL,
    ->     message     VARCHAR(255) NOT NULL,
    ->     type        VARCHAR(50),
    ->     sent_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     read_status BOOLEAN DEFAULT FALSE,
    ->     FOREIGN KEY (user_id) REFERENCES Users(user_id) ON DELETE CASCADE
    -> );
```



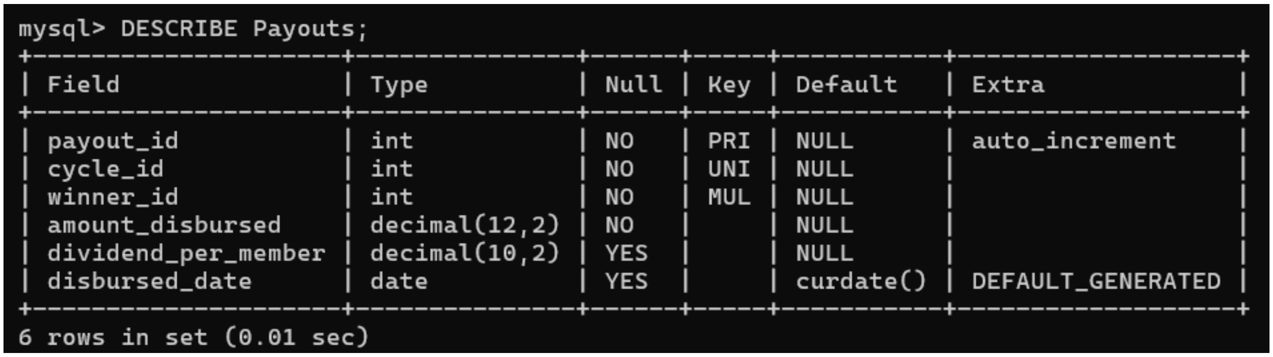
```
mysql> DELIMITER //
mysql>
mysql> -- Prevent a member who already won a payout in this group from bidding again
mysql> CREATE TRIGGER trg_prevent_repeat_bid
    -> BEFORE INSERT ON Bids
    -> FOR EACH ROW
    -> BEGIN
    ->     DECLARE already_won BOOLEAN;
    ->     SELECT gm.has_won_payout INTO already_won
    ->     FROM GroupMembers gm
    ->     JOIN MonthlyCycles mc ON mc.group_id = gm.group_id
    ->     WHERE mc.cycle_id = NEW.cycle_id AND gm.user_id = NEW.user_id;
    ->
    ->     IF already_won = TRUE THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Member has already won a payout in this group and cannot bid again';
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> -- Auto-flag a defaulter when a payment is marked 'defaulted'
mysql> CREATE TRIGGER trg_flag_defaulter
    -> AFTER UPDATE ON Payments
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.status = 'defaulted' AND OLD.status <> 'defaulted' THEN
    ->         INSERT INTO Defaulters (user_id, group_id, cycle_id, amount_due, penalty_applied)
    ->         SELECT NEW.user_id, mc.group_id, NEW.cycle_id, NEW.amount_paid, NEW.penalty_amount
    ->         FROM MonthlyCycles mc WHERE mc.cycle_id = NEW.cycle_id;
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> -- Lock the winner's flag once a payout is created for a cycle
mysql> CREATE TRIGGER trg_lock_winner
    -> AFTER INSERT ON Payouts
    -> FOR EACH ROW
    -> BEGIN
    ->     UPDATE GroupMembers gm
    ->     JOIN MonthlyCycles mc ON mc.group_id = gm.group_id
    ->     SET gm.has_won_payout = TRUE
    ->     WHERE mc.cycle_id = NEW.cycle_id AND gm.user_id = NEW.winner_id;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```



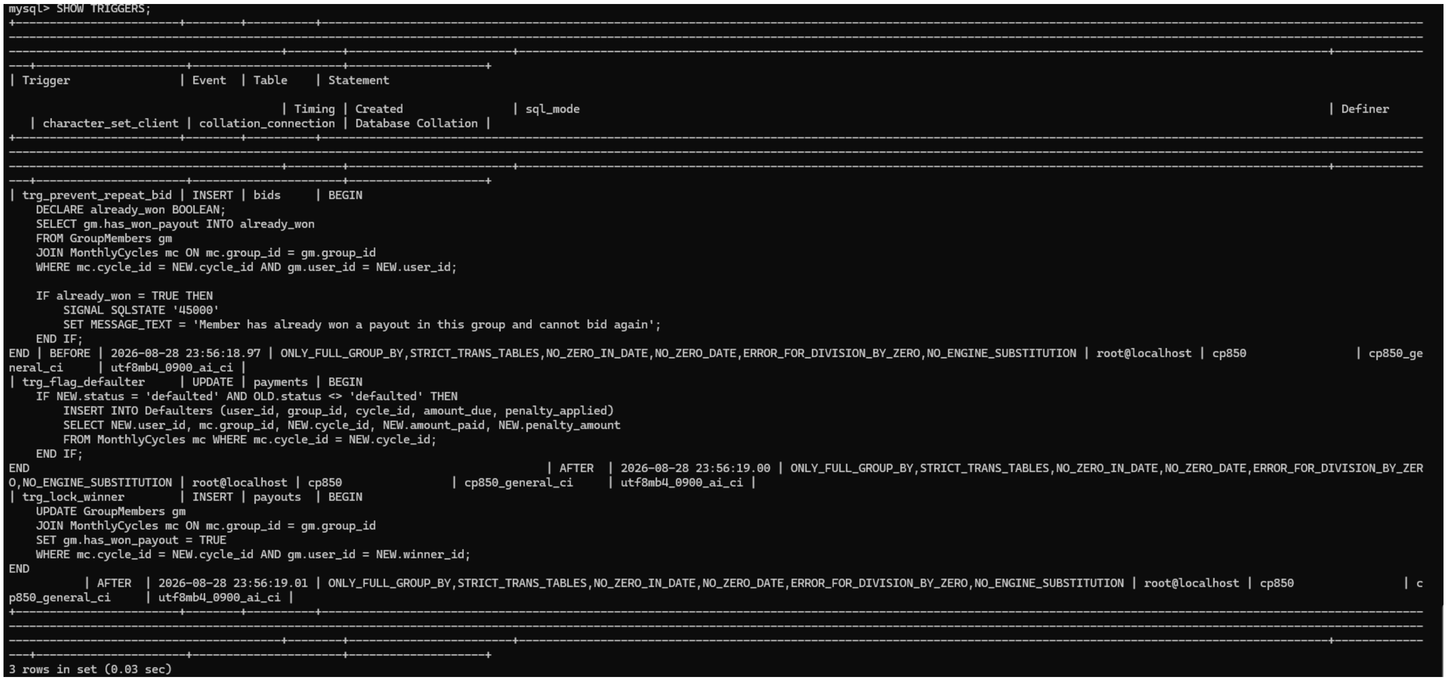
```
mysql> SHOW TABLES;
+----------------------+
| Tables_in_chitvault |
+----------------------+
| bids                 |
| chitgroups           |
| defaulters           |
| groupmembers         |
| kycdocuments         |
| monthlycycles        |
| notifications        |
| payments             |
| payouts              |
| users                |
+----------------------+
10 rows in set (0.04 sec)
```



```
mysql> DESCRIBE ChitGroups;
+---------------------+------------------------------------------------------+------+-----+-------------------+-------------------+
| Field               | Type                                                 | Null | Key | Default           | Extra             |
+---------------------+------------------------------------------------------+------+-----+-------------------+-------------------+
| group_id            | int                                                  | NO   | PRI | NULL              | auto_increment    |
| group_name          | varchar(100)                                         | NO   |     | NULL              |                   |
| organizer_id        | int                                                  | NO   | MUL | NULL              |                   |
| total_amount        | decimal(12,2)                                        | NO   |     | NULL              |                   |
| monthly_installment | decimal(10,2)                                        | NO   |     | NULL              |                   |
| duration_months     | int                                                  | NO   |     | NULL              |                   |
| max_members         | int                                                  | NO   |     | NULL              |                   |
| start_date          | date                                                 | NO   |     | NULL              |                   |
| status              | enum('upcoming','active','completed','cancelled')    | YES  |     | upcoming          |                   |
| created_at          | timestamp                                            | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------------+------------------------------------------------------+------+-----+-------------------+-------------------+
10 rows in set (0.03 sec)
```



```
mysql> DESCRIBE Payouts;
+---------------------+---------------+------+-----+-----------+-------------------+
| Field               | Type          | Null | Key | Default   | Extra             |
+---------------------+---------------+------+-----+-----------+-------------------+
| payout_id           | int           | NO   | PRI | NULL      | auto_increment    |
| cycle_id            | int           | NO   | UNI | NULL      |                   |
| winner_id           | int           | NO   | MUL | NULL      |                   |
| amount_disbursed    | decimal(12,2) | NO   |     | NULL      |                   |
| dividend_per_member | decimal(10,2) | YES  |     | NULL      |                   |
| disbursed_date      | date          | YES  |     | curdate() | DEFAULT_GENERATED |
+---------------------+---------------+------+-----+-----------+-------------------+
6 rows in set (0.01 sec)
```



```
mysql> SHOW TRIGGERS;
+-----------------------+--------+----------+----------------------------------------------------------------------------------------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------+--------+------------------------+------------------------------------------------------------------------------------------------------------------------------+------------
---+----------------------+----------------------+--------------------+
| Trigger               | Event  | Table    | Statement

                                    | Timing | Created                | sql_mode                                                                                                                     | Definer
   | character_set_client | collation_connection | Database Collation |
+-----------------------+--------+----------+----------------------------------------------------------------------------------------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------------+--------+------------------------+------------------------------------------------------------------------------------------------------------------------------+------------
---+----------------------+----------------------+--------------------+
| trg_prevent_repeat_bid | INSERT | bids     | BEGIN
    DECLARE already_won BOOLEAN;
    SELECT gm.has_won_payout INTO already_won
    FROM GroupMembers gm
    JOIN MonthlyCycles mc ON mc.group_id = gm.group_id
    WHERE mc.cycle_id = NEW.cycle_id AND gm.user_id = NEW.user_id;

    IF already_won = TRUE THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Member has already won a payout in this group and cannot bid again';
    END IF;
END | BEFORE | 2026-08-28 23:56:18.97 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850                | cp850_ge
neral_ci     | utf8mb4_0900_ai_ci |
| trg_flag_defaulter     | UPDATE | payments | BEGIN
    IF NEW.status = 'defaulted' AND OLD.status <> 'defaulted' THEN
        INSERT INTO Defaulters (user_id, group_id, cycle_id, amount_due, penalty_applied)
        SELECT NEW.user_id, mc.group_id, NEW.cycle_id, NEW.amount_paid, NEW.penalty_amount
        FROM MonthlyCycles mc WHERE mc.cycle_id = NEW.cycle_id;
    END IF;
END                                                                                    | AFTER  | 2026-08-28 23:56:19.00 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZER
O,NO_ENGINE_SUBSTITUTION | root@localhost | cp850                | cp850_general_ci     | utf8mb4_0900_ai_ci |
| trg_lock_winner        | INSERT | payouts  | BEGIN
    UPDATE GroupMembers gm
    JOIN MonthlyCycles mc ON mc.group_id = gm.group_id
    SET gm.has_won_payout = TRUE
    WHERE mc.cycle_id = NEW.cycle_id AND gm.user_id = NEW.winner_id;
END
           | AFTER  | 2026-08-28 23:56:19.01 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850                | c
p850_general_ci     | utf8mb4_0900_ai_ci |
+-----------------------+--------+----------+----------------------------------------------------------------------------------------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
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---+----------------------+----------------------+--------------------+
3 rows in set (0.03 sec)
```